LNP™ THERMOCOMP™ COMPOUND D351

DESCRIPTION

LNP THERMOCOMP D351 (experimental grade name as EXTC8143) is a compound based on Polycarbonate resin containing Glass Fiber, Flame Retardant. Added features of this material include: High modulus, good flatness, good ductility, Non-Brominated & Non-Chlorinated Flame Retardant.

TYPICAL PROPERTY VALUES

Revision 20191217

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
MECHANICAL			
Tensile Stress, brk, Type I, 5 mm/min	120	MPa	ASTM D 638
Tensile Strain, brk, Type I, 5 mm/min	2.4	%	ASTM D 638
Tensile Modulus, 5 mm/min	9040	MPa	ASTM D 638
Flexural Stress, brk, 1.3 mm/min, 50 mm span	179	MPa	ASTM D 790
Flexural Modulus, 1.3 mm/min, 50 mm span	8390	MPa	ASTM D 790
Tensile Stress, break, 5 mm/min	125	MPa	ISO 527
Tensile Modulus, 1 mm/min	9080	MPa	ISO 527
Flexural Stress, yield, 2 mm/min	169	MPa	ISO 178
Flexural Modulus, 2 mm/min	7520	MPa	ISO 178
IMPACT			
Izod Impact, unnotched, 23°C	508	J/m	ASTM D 4812
Izod Impact, notched, 23°C	150	J/m	ASTM D 256
Instrumented Impact Energy @ peak, 23°C	24	J	ASTM D 3763
Charpy 23°C, V-notch Edgew 80*10*4 sp=62mm	15	kJ/m²	ISO 179/1eA
Charpy 23°C, Unnotch Edgew 80*10*4 sp=62mm	39	kJ/m²	ISO 179/1eU
THERMAL			
HDT, 1.82 MPa, 3.2mm, unannealed	116	°C	ASTM D 648
CTE, -40°C to 40°C, flow	2.02E-05	1/°C	ASTM E 831
CTE, -40°C to 40°C, xflow	6.74E-05	1/°C	ASTM E 831
Relative Temp Index, Elec (1)	80	°C	UL 746B
Relative Temp Index, Mech w/impact (1)	80	°C	UL 746B
Relative Temp Index, Mech w/o impact (1)	80	°C	UL 746B
PHYSICAL			
Density	1.425	g/cm³	ASTM D 792
Mold Shrinkage, flow, 24 hrs	0.1 – 0.3	%	ASTM D 955
Mold Shrinkage, xflow, 24 hrs	0.1 – 0.3	%	ASTM D 955
Melt Flow Rate, 300°C/2.16 kgf	25.2	g/10 min	ASTM D 1238
Melt Volume Rate, MVR at 300°C/2.16 kg	20	cm³/10 min	ISO 1133
FLAME CHARACTERISTICS (1)			
UL Yellow Card Link	E207780-101265024	-	-
UL Recognized, 94-5VA Flame Class Rating	≥3	mm	UL 94
UL Recognized, 94V-0 Flame Class Rating	≥0.8	mm	UL 94
UL Recognized, 94V-1 Flame Class Rating	≥0.6	mm	UL 94
INJECTION MOLDING			

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
Drying Temperature	110	°C	
Drying Time	3 – 6	hrs	
Maximum Moisture Content	0.02	%	
Melt Temperature	285 – 310	°C	
Nozzle Temperature	285 – 305	°C	
Front - Zone 3 Temperature	280 – 300	°C	
Middle - Zone 2 Temperature	270 – 290	°C	
Rear - Zone 1 Temperature	260 – 280	°C	
Mold Temperature	80 – 110	°C	
Back Pressure	0.1 – 0.3	MPa	
Screw Speed	50 – 90	rpm	

(1) UL Ratings shown on the technical datasheet might not cover the full range of thicknesses and colors. For details, please see the UL Yellow Card.

DISCLAIMER